Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

BIOLOGY **0610/42**

Paper 4 Theory (Extended) **March 2017**

MARK SCHEME

Maximum Mark: 80

Published

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Mark schemes will use these abbreviations

- ; separates marking points
- / alternatives
- **I** ignore
- **R** reject
- **A** accept (for answers correctly cued by the question, or guidance for examiners)
- AW alternative wording (where responses vary more than usual)
- AVP any valid point
- ecf credit a correct statement / calculation that follows a previous wrong response
- **ora** or reverse argument
- ( ) the word / phrase in brackets is not required, but sets the context
- underline actual word given must be used by candidate (grammatical variants excepted)
- max indicates the maximum number of marks that can be given

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Question	Answer	Marks	Guidance
1(a)(i)	**L** – atrioventricular valve **;** **M** – septum **;** **O** – semi-lunar valve **;**	**3**	
1(a)(ii)	**N / P ;** **J / K ;** **J ;**	**3**	
1(b)(i)	**1** blood from pulmonary vein / K, enters left atrium **;** **2** atria contract **;** **3** atrioventricular valve / L, <u>opens</u> due to pressure from blood **;** **4** blood forced into left ventricle **;** **5** ventricle contract **;** **6** atrioventricular valves / L, shut to prevent blood entering atrium **;** **7** semi-lunar valves / O, open **;** **8** blood forced into, aorta / J **;** **9** AVP**;**	**5**	
1(b)(ii)	left ventricle wall contains more muscle **;** left ventricle pumps blood further **;** left ventricle has to overcome more resistance **;** left ventricle pumps blood at higher pressure **;**	**2**	

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Question	Answer	Marks	Guidance
2(a)	no nucleus **;** cell wall **;** loop of DNA **;** AVP **;;**	**2**	
2(b)(i)	overall increase in number of cases of MRSA **;** largest increase, between 2004–2005 / exponential **;** data quote including the number of cases and the year / data manipulation **;**	**2**	
2(b)(ii)	**1** correct ref to mutation of bacteria **;** **2** variation in ability of bacteria to survive antibiotic treatment **;** **3** bacteria with no / little resistance, die **;** **4** bacteria with resistance, survive and breed **;** **5** passing on resistant allele **;** **6** ref to natural selection **;** **7** AVP **;** e.g. ref to strengthening of cell wall	**4**	
2(c)	more responsible use of antibiotics **;** improved, detection / screening to avoid spread **;** ref to improved cleanliness **;** isolating infected patients **;** development of new antibiotics / treatment **;**	**2**	

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>3(a)</td><td><table><tr><td>part of the eye</td><td>function</td></tr><tr><td>rod cells</td><td>night vision / detects low light ;</td></tr><tr><td>cone cells</td><td>colour vision ;</td></tr><tr><td>sensory neurone</td><td>transmits nerve impulses to brain ;</td></tr></table></td><td>3</td><td>1 mark for each correct row</td></tr>
<tr><td>3(b)</td><td>1 more rod cells than cone cells in the retina ;
2 ref to uneven distribution of rod cells either side of fovea ;
3 no rod cells and no cone cells at blind spot ;
4 optic nerve enters / leaves retina at blind spot ;
5 only cone cells at the fovea / no rod cells at the fovea ;
6 maximum number of cone cells are at the, fovea / 0 degrees ;
7 maximum number of rod cells at 20–21 degrees ;
8 data quote include units ;
9 AVP ;
10 AVP ;</td><td>5</td><td></td></tr>
<tr><td>3(c)</td><td>more males affected than females / ora ;
only females are carriers / males are affected or not ;</td><td>2</td><td></td></tr>
<tr><td>3(d)</td><td><table><tr><td><u>correct gametes ;</u>
<u>correct offspring genotypes ;</u>
<u>correct offspring phenotypes ;</u>

<u>correct percentage ;</u></td><td>X^B,Y +X^b, X^b ;
X^BX^b, X^BX^b, X^bY, X^bY;
carrier female, carrier female, colour-blind male, colour-blind male ;
50 % ;</td></tr></table></td><td>4</td><td>offspring phenotype must be linked to the correct offspring genotype</td></tr>
</table>

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Question	Answer	Marks	Guidance
4(a)	carbon dioxide ; light energy ; chlorophyll ;	**2**	
4(b)	(2 ÷ 13) × 100 ; 15(%) ;	**2**	
4(c)(i)	increased rate of transpiration ; greater concentration of water vapour inside the leaf than outside ; more water vapour diffuses out of the leaf ; through stomata ; more water is drawn up through xylem / transpiration pull ;	**3**	
4(c)(ii)	by osmosis ; the soil has a higher <u>water potential</u> than the root cells ; water moves from an area of higher water potential to lower water potential ; across a partially permeable membrane ; ref to root hair cell ;	**3**	**A** down a water potential gradient
4(d)	**1** loss of habitat ; **2** population decrease / migration ; **3** extinction / endangerment, of species ; **4** loss of biodiversity ; **5** less food ; **6** disruption of, food chains / food webs ;	**4**	

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Question	Answer	Marks	Guidance
5(a)(i)	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ **;;**	**2**	
5(a)(ii)	liver **;**	**1**	
5(b)	correct ref to active site **;** enzyme must be complementary shape to, substrate / alcohol **;** to make enzyme – substrate complex / to allow substrate to bind to enzyme **;** ref to only fits one substrate / specific to one substrate **;**	**3**	**A** 'lock and key'
5(c)(i)	increased <u>kinetic</u> energy **;** molecules move faster **;** increased frequency of collisions **;** increased number of successful collisions **;**	**3**	
5(c)(ii)	pH **;**	**1**	
5(d)(i)	length of DNA **;** that codes for a protein **;**	**2**	
5(d)(ii)	mRNA passes through ribosomes **;** ribosomes assemble amino acids into proteins **;** order of amino acids is determined by the sequence of <u>bases</u> in mRNA **;** AVP **;**	**2**	

ROOT 19

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>6(a)</td><td>
<table>
<tr><th>enzyme</th><th>substrate</th><th>product / s</th><th>location of enzyme production</th><th></th></tr>
<tr><td>(salivary) amylase</td><td>starch</td><td>maltose</td><td>salivary glands</td><td>;</td></tr>
<tr><td>maltase</td><td>maltose</td><td>glucose</td><td>small intestinal wall</td><td>;</td></tr>
<tr><td><u>pepsin</u></td><td>protein</td><td>amino acids</td><td>stomach (wall)</td><td>;</td></tr>
<tr><td><u>trypsin</u></td><td>protein</td><td>amino acids</td><td>small intestinal (wall)</td><td>;</td></tr>
<tr><td>lipase</td><td>fats</td><td>fatty acids and glycerol</td><td>pancreas / small intestinal wall</td><td>;</td></tr>
</table>
</td><td>5</td><td>A polypeptides for protein

A peptides for protein</td></tr>
<tr><td>6(b)</td><td><u>emulsi</u>fication ;
increased surface area of fat globules ;
faster, digestion / break down of fat by enzymes ;
by lipase / to fatty acids <u>and</u> glycerol ;
neutralises (stomach) acid ;</td><td>3</td><td></td></tr>
<tr><td>6(c)</td><td>the movement of small food molecules and ions ;
through the <u>wall</u> of the intestine ;
into the blood ;</td><td>3</td><td></td></tr>
</table>

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Question	Answer	Marks	Guidance
6(d)	marasmus / kwashiorkor **;**	**1**	
6(e)	reduces, calorie / energy intake **;** reduces obesity **;** reduces chances of CHD **;** AVP **;;**	**3**	

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